CHORUS-SGH-1 Contributors Agreement

differential-harness · IP, inventorship & credit allocation

CHORUS Research Program

Root	/home/joeblack/Documents/differential-harness

Joseph Black

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docs/legal/CHORUS*SGH1*Contributors_Agreement.md

CHORUS-SGH-1 Contributors Agreement

Project: differential-harness — CHORUS-Skid **SGH-1** + **AEH-1**

Program: CHORUS Research Program

Principal lead: Joseph Black

Effective template date: 2026-06-01

Repository: `differential-harness` (GitHub / private distribution as applicable)

> **Disclaimer:** This document is a **project governance template**, not legal advice. Before signing, funding, filing patents, or taking investment, have a qualified attorney review it for your jurisdiction and entity structure.

1. Purpose

This agreement defines how people contribute to the **CHORUS-SGH-1** bench skid (pressure-retarded osmosis + anthropogenic brine pairing + acoustic energy harvest / ultrasonic assist + shared DAQ), how **credit** is recorded, and how **intellectual property** is handled relative to open-source code and separate patent / commercial tracks.

2. Definitions

Term	Meaning
Project	The `differential-harness` repository and related hardware, simulation, DAQ, documentation, and research outputs for SGH-1 / AEH-1.
Principal lead	Joseph Black — originator of the integrated CHORUS-SGH-1 architecture and default coordinator for filings, publications, and external outreach unless reassigned in writing.
Contributor	Any individual or organization that merges work, supplies data under this agreement, or signs below.
Substantive contribution	Original work that is more than typo fixes: simulation modules, CAD parts, bench protocols, experimental datasets, claim-relevant design choices, or major documentation that advances reduction-to-practice.

| **Entity** | The legal person that will own filed patents and commercial licenses (e.g. **CHORUS Research Program**, an LLC, or university — fill in before signatures). |

3. What you may contribute

Contributions include, without limitation:

- **Software:** Python simulation (`simulation/`), DAQ (`daq/`), notebooks, paper pipeline scripts
- **Hardware:** OpenSCAD / STL, BOM lines, wiring, build and test protocols
- **Research:** Figures, calibration JSON, bench CSVs, manuscript sections
- **Operations:** Lab time, machining, sensors, hosting, grant administration

Each contribution should be tied to a **dated record**: git commit, inventor-notebook entry (see `docs/INVENTOR_NOTEBOOK.md`), and/or signed exhibit list (Appendix A).

4. Open-source license (code & docs)

Unless marked otherwise in the file header:

- **Repository code and docs** are contributed under the **MIT License** (or the license stated in the repo root at time of merge).
- You represent that you have the right to contribute the material and that it does not knowingly violate third-party rights.
- **Pre-existing IP:** List any background patents, designs, or code in **Appendix B**. Background IP stays yours; you grant the Project a non-exclusive license to use it only as embedded in your contribution.

5. Credit tiers (how we split "credit")

Credit is **not one-size-fits-all**. Different outputs use different rules.

5.1 Tier summary

Tier	Typical role	Where credited	How earned
A — Principal lead	Architecture, program direction	Cover page, patent lead contact, "corresponding author" on flagship paper	Joseph Black (unless transferred by written amendment)
B — Co-inventor	Conception / reduction to practice on **patent claims**	USPTO inventor list on filed applications	Substantive contribution to **at least one claim** (US inventorship rules); see §6

| **C — Publication author** | Analysis, writing, experiments | Research PDF / journal byline | Substantive work on **that publication** (ICMJE-style adapted); order negotiated before submission |

| **D — Repository contributor** | Merged PRs, CAD, tooling | `CONTRIBUTORS.md`, release notes, optional “Contributors” section in compendium PDF | Merged contribution + name in Appendix A |

| **E — Acknowledgment** | Funding, facilities, review | “Acknowledgments” in papers and decks | No IP claim; name only |

5.2 Publication authorship (default policy)

For the flagship report *CHORUS-SGH-1: Brine-Gradient Power on a Bench Skid* and derivative papers:

- **First / corresponding author:** Principal lead unless all co-authors agree otherwise in writing before submission.
- **Co-authors:** Must (a) contribute to design, analysis, or drafting, and (b) approve the final manuscript.
- **Author order** after the lead: by **negotiated** contribution (weighted: new physics/analysis > hardware build > software plumbing > editorial).
- **Guest / honorary authorship is not permitted.**

5.3 Engineering & repo attribution

- **CAD parts:** Credit in `hardware/COMPONENT_INDEX.md` and commit history; major new subsystems get a line in release notes.
- **Simulation:** Module docstring + `exports/paper_experiments.json` provenance when applicable.
- **Bench data:** CSV metadata (`data/bench/*.meta.json`) lists operator, date, and contributors.

6. Patent inventorship & ownership

Patent credit follows **US law (and equivalent rules elsewhere):** inventors are natural persons who contributed to the **conception** of at least one claim, not sponsors or “project managers” alone.

6.1 Default claim buckets (SGH-1)

Align contributions to these focus areas from `docs/SGH1`*PATENT*`AND_DEPLOYMENT.md`:

- Integrated skid: PRO + CHOR plenum + AEH panel + shared DAQ bus
- Anthropogenic brine + low-sal feed pairing on modular manifold
- $\Delta P/\Delta\pi$ feedback using dual conductivity sensors
- Dual-mode acoustic: harvest + ultrasonic CP assist

Co-inventor status requires documented input to one or more of these **before** public disclosure (conference, arXiv, open repo without NDA, etc.).

6.2 Ownership & assignment

• Filed patents are owned by **Entity:** ______________________________

• Contributors who qualify as inventors **assign** their patent rights to Entity (or agree to cooperate in filing) via signature below.

• Entity agrees to **credit all named inventors** on filings and to negotiate revenue share per §7.

6.3 Disclosure duty

Contributors must **promptly disclose** prior publications, patents, or public talks that might be prior art. Do not commit claim-critical ideas to public repos until provisional strategy is confirmed with the lead.

7. Commercial & revenue split (if monetized)

If the Project generates **licensing revenue, grants allocated to commercialization, or acquisition proceeds** tied to SGH-1 IP, default **negotiation starting point** (adjust before signing):

Stakeholder pool	Suggested share of net revenue*	Notes
Principal lead	**40%**	Program risk, architecture, coordination
Co-inventors (pool)	**35%**	Split **equally** among named inventors on the issued patent(s)
Key contributors (non-inventor pool)	**15%**	Split by written **Contribution Points** (Appendix C)
Entity / reserve	**10%**	Legal, filing fees, future R&D

\ Net* = gross receipts minus direct licensing costs, filing maintenance, and agreed entity overhead (define in entity operating agreement).

Contribution Points (Appendix C) — example weights (lead assigns at milestone review):

Activity	Points
New claim-relevant subsystem conceived + documented	10
Reduction-to-practice bench result (positive power or key metric)	8
Major simulation / sizing module	5

| Production CAD + BOM for build | 4 |

| DAQ / automation enabling publishable dataset | 3 |

| Sustained review + integration (6+ months) | 2 |

Pools are **renegotiated** when a new co-inventor joins or at each provisional / non-provisional filing.

8. Confidentiality & NDAs

• **Public repo** content is not confidential once merged to default branch.

• **Bench data, investor decks, and pre-filing designs** may be shared under a separate NDA — do not forward without lead approval.

• Outreach package per patent playbook: 1-pager + design report + data **under NDA** until counsel approves otherwise.

9. Representations

Each contributor represents that:

• Their contribution is their own work (or properly licensed).

• They will not intentionally introduce malware or fabricated data.

• They will abide by lab safety and export control rules applicable to their site.

10. Term & withdrawal

• This agreement applies from the **signature date** forward to contributions made while participating.

• Withdrawal: written notice to the lead; **prior licenses and assignments survive** for work already merged or filed.

• Removal from inventor list after filing requires counsel — do not unilaterally edit patent records.

11. Dispute resolution

• **Good-faith negotiation** between contributor and principal lead (14 days).

• **Mediation** in ____________________ (city/state) if unresolved.

• Litigation only as a last resort; prevailing-party fees only if required by entity bylaws.

12. Signatures

By signing, you agree to this template as amended by **Appendix A–C** for your role.

Role	Printed name	Entity (if any)	Signature	Date
Principal lead	Joseph Black	CHORUS Research Program		
Contributor				
Contributor				
Entity representative				

Appendix A — Contribution log (exhibit)

Date	Contributor	Description (commit / notebook / part)	Tier (B–E)

Appendix B — Background IP disclosure

Contributor	Description of pre-existing IP	License to Project

Appendix C — Contribution points ledger

Contributor	Points	Milestone / notes

Quick reference: “How do we split credit?”

Question	Answer
GitHub / repo fame?	Tier D — merged work + `CONTRIBUTORS.md` + Appendix A
Paper byline?	Tier C — negotiate before submit; lead is default corresponding author
Patent name on filing?	Tier B only — real inventorship per §6; no “gift” inventors
Money from a license?	§7 pools — inventors split 35%; lead 40%; points pool 15%
Who decides disputes?	Lead first, then mediation (§11)

Generated from project template · Build PDF: `./scripts/build`*`contributors`*`agreement.sh`

Export summaries

exports/chorus_results.json

Key	Value
claims.civilization	Full CHORUS column on coastal km²
claims.safest	Blue energy (RED) at estuaries
claims.smartest	PV + hydrovoltaic evaporative coupling
claims.strangest	Telluric Storm Coupling conductance network
constants.C_river	20.0
constants.C_sea	600.0
constants.T_K	298.15
constants.n_pairs	50
generated_at	2026-06-01T05:41:40.929415+00:00
results.E_N_mV	87.38553283478494
results.P_blue_W_m2	15.0
results.P_glob_GW	0.6
results.P_mfc_uW_m2	36.74903137795773
results.P_mix_max_MW	1437.795077169385
results.P_pv_W_m2	187.90754624152774
results.P_pv_gain_W_m2	5.621546241527739
results.R_int_ohm_m2	0.3181763062008032
results.V_stack_V	4.369276641739247
results.column_MW_median	22.758638714734108
results.column_MW_p10	19.625723332027405
results.column_MW_p90	26.464372583645808
title	CHORUS Physics Proof

exports/openscad_audit.json

Key	Value
aeh_panel.cavity_d_mm	35
aeh_panel.estimated_cell_pitch_x_mm	41.142857142857146

Key	Value
aeh_panel.estimated_cell_pitch_z_mm	80.0
aeh_panel.neck_d_mm	8
aeh_panel.panel_h_mm	400
aeh_panel.panel_t_mm	28
aeh_panel.panel_w_mm	288.0
aeh_panel.resonator_cells	24
assembly_files	[list len=2]
generated_constants.A_mem_m2	0.72
generated_constants.P_target_W	10.0
generated_constants.active_h	300.0
generated_constants.active_w	200.0
generated_constants.bolt_circle	240.0
generated_constants.bore_id_mm	260.0
generated_constants.delta_P_bar	34.58
generated_constants.frame_H	900.0
generated_constants.frame_L	664.0
generated_constants.frame_W	480.0
generated_constants.housing_len	264.0
generated_constants.housing_od	280.0
part_count	23
parts	[list len=23]

exports/paper_experiments.json

Key	Value
column_monte_carlo.N	8000
column_monte_carlo.column_MW_median	22.905785810200392
column_monte_carlo.column_MW_p10	19.66848061978812
column_monte_carlo.column_MW_p90	26.692656379993203
column_monte_carlo.layers.blue_energy.median_MW	0.26325373174450273

Key	Value
column_monte_carlo.layers.blue_energy.p10_MW	0.18065760631962177
column_monte_carlo.layers.blue_energy.p90_MW	0.3863177178459722
column_monte_carlo.layers.blue_energy.share_of_median_column_pct	1.1492892403947597
column_monte_carlo.layers.meg.median_MW	0.011045001778810973
column_monte_carlo.layers.meg.p10_MW	0.005453996352434218
column_monte_carlo.layers.meg.p90_MW	0.022423577362723508
column_monte_carlo.layers.meg.share_of_median_column_pct	0.04821926595459745
column_monte_carlo.layers.pv_hydro.median_MW	22.6149731808767
column_monte_carlo.layers.pv_hydro.p10_MW	19.373662625586938
column_monte_carlo.layers.pv_hydro.p90_MW	26.38351968866515
column_monte_carlo.layers.pv_hydro.share_of_median_column_pct	98.7803966267152
column_monte_carlo.layers.smfc.median_MW	2.21121497379417e-05
column_monte_carlo.layers.smfc.p10_MW	1.2154259652746951e-05
column_monte_carlo.layers.smfc.p90_MW	4.0352984257319246e-05
column_monte_carlo.layers.smfc.share_of_median_column_pct	9.653521569251175e-05
column_monte_carlo.parcel_area_m2	1000000.0
estuary_RED.E_N_mV	87.38553283617222
estuary_RED.P_max_W_m2	15.0
estuary_RED.P_mix_ceiling_MW	1437.7950771428862

exports/real_world_calibration.json

Key	Value
case_studies	[list len=5]
commercial_pro_6p3.A_m2_for_10W	1.5873015873015874
commercial_pro_6p3.P_W_m2	6.3
commercial_pro_6p3.P_est_W	10.0
mixing_energy.literature_WA_table	0.14
mixing_energy.seawater_brine_kWh_m3	0.8263190098522333
perth_parasitic_fraction.daily_desal_MWh	504.0

Key	Value
perth_parasitic_fraction.daily_pro_kWh_at_10W	0.24
perth_parasitic_fraction.fraction_of_plant_energy_pct	4.761904761904762e-05
perth_parasitic_fraction.note	Single bench skid; scaled fleet would multiply P_pro
perth_sidestream_pro.P_W_Lp1e12	1.2162897261298127
perth_sidestream_pro.c_draw	1200.0
perth_sidestream_pro.c_feed	5.0
perth_sidestream_pro.delta_pi_MPa	5.924707300640513
perth_sidestream_pro.label	Perth-class brine + WW
perth_sidestream_pro.mixing_kWh_m3	0.8263190098522333
statkraft_estuary.E_N_mV	87.38553283617222
statkraft_estuary.P_max_W_m2	15.00831633070309
statkraft_estuary.V_stack_V	4.369276641808611
statkraft_estuary.delta_pi_MPa	2.8755901542857716
tsc_baseline.P_dissipated_W	0.00014516129032258055
tsc_baseline.currents_A	[list len=3]
tsc_baseline.psi_a_V	120.96774193548383
tsc_baseline.psi_s_V	145.16129032258058

exports/sgh1_design.json

Key	Value
claims.civilization	Co-located desal brine + effluent + CHORUS skid
claims.safest	PRO salinity-gradient core
claims.smartest	Ultrasonic CP gain + PRO
claims.strangest	TSC + AEH urban panel
sizing.A_mem_m2	0.72
sizing.P_density_W_m2	8.0
sizing.P_target_W	10.0
sizing.Q_feed_L_min	0.14939186642920496
sizing.active_height_mm	300.0

Key	Value
sizing.active_width_mm	200.0
sizing.bolt_pattern_mm	240.0
sizing.c_draw	1400.0
sizing.c_feed	5.0
sizing.delta_P_star_MPa	3.4581450562315963
sizing.delta_P_star_bar	34.581450562315965
sizing.delta_pi_MPa	6.916290112463193
title	CHORUS-Skid SGH-1 Design Export

exports/sgh1_pi_groups.json

Key	Value
L_p_required_for_target	6.033236600629109e-12
P_sim_W	1.65748513806955
P_target_W	10.0
Pi1_delta_P_over_delta_pi	0.5
Pi2_Jw_scaled	4.158211788877279e-11
Pi3_Pe_order	0.6916290112463191
Pi4_area_law	0.576
Pi5_sim_over_target	0.165748513806955

exports/sgh1_sizing.json

Key	Value
A_mem_m2	0.72
P_density_W_m2	8.0
P_target_W	10.0
Q_feed_L_min	0.14939186642920496
active_height_mm	300.0
active_width_mm	200.0
bolt_pattern_mm	240.0

Key	Value
c_draw	1400.0

exports/symbolic_checks.json

Key	Value
E_N_mV_brine_pair	144.7647208666358
available	True
delta_pi_MPa_brine_pair	6.915905288999999
kim_baker_at_half_delta_pi	dpi/2
kim_baker_critical_points	[list len=1]
latex_E_N	\frac{R T \log{\left(\frac{c_{h}}{c_{l}} \right)}}{F z}
latex_delta_pi	R T i \left(c_{h} - c_{l}\right)